1. Introduction

1.1 Project Overview

Laboratory equipment plays an important role in engineering education. Managing equipment manually using paper records or spreadsheets is time-consuming and may lead to errors such as duplicate bookings, missing records, and inaccurate inventory tracking.

The **University Lab Equipment Management System (LabTrack)** is a web-based application developed using **Laravel 12** and **Oracle Database 11g XE**. The system digitalizes laboratory operations by allowing users to manage equipment, submit booking requests, borrow and return equipment, monitor fines, and generate reports. It also supports role-based access for **Lab Assistants, Teachers, and Students**, ensuring that each user can access only the features relevant to their responsibilities.

1.2 Objectives

The main objectives of this project are:

- Develop a centralized laboratory equipment management system.
- Maintain accurate equipment inventory records.
- Automate equipment booking and borrowing processes.
- Manage overdue returns and fine records.
- Generate summary reports for administrative purposes.
- Ensure secure access through role-based authentication.

1.3 Features

The system is designed for use within a university laboratory environment.

The major features include:

- Equipment Management
- Booking Management
- Borrow and Return Management
- Fine Management
- Student and Teacher Management
- Report Generation
- Role-Based Access Control

The system focuses on improving laboratory resource management while reducing manual work and maintaining data consistency through Oracle Database.

2. System Analysis

2.1 Existing System

Many university laboratories still manage equipment manually using paper registers or spreadsheets. This approach has several limitations:

- Time-consuming record management.
- Difficulty in tracking available equipment.
- Risk of duplicate or lost records.
- Manual calculation of fines.
- Limited reporting and monitoring facilities.

These issues reduce efficiency and make equipment management more difficult.

2.2 Proposed System

The proposed **University Lab Equipment Management System** provides a centralized web-based platform to manage laboratory equipment efficiently.

The system enables users to:

- View available equipment.
- Submit booking requests.
- Borrow and return equipment.
- Track fine records.
- Generate reports.
- Manage students and teachers.
- Maintain accurate inventory records.

The system also uses role-based access control to ensure secure and authorized operations.

The system provides the following major functions:

- User Login
- Equipment Management
- Booking Management
- Borrow Management
- Fine Management
- Student Management
- Teacher Management
- Report Generation

3. System Design

3.1 System Architecture

The University Lab Equipment Management System follows the **Model-View-Controller (MVC)** architecture provided by the Laravel framework.

The system consists of three main components:

- **Frontend:** Developed using HTML, CSS, Bootstrap 5, and Blade templates.
- **Application Layer:** Laravel controllers handle business logic, validation, and user requests.
- **Database Layer:** Oracle Database 11g XE stores all project data, including users, equipment, bookings, borrow records, fines, and reports.

3.2 System Workflow

The overall workflow of the system is shown below:

1. User logs into the system.
2. Student submits an equipment booking request.
3. Teacher reviews and approves or rejects the request.
4. Lab Assistant issues the equipment.
5. Student returns the equipment.
6. If the return is overdue, a fine is generated.
7. Reports are updated automatically.

4. Database Design

4.1 Database Tables

The system uses **Oracle Database 11g XE** to store and manage all application data. A relational database model is used to maintain consistency and minimize data redundancy.

The database consists of the following tables:

Table	Purpose
Users	Stores student, teacher, and lab assistant information.
Labs	Stores laboratory information.
Categories	Stores equipment categories.
Equipment	Stores laboratory equipment details.

Booking Requests	Stores equipment booking requests.
Borrow Records	Stores borrowing and return information.
Damage Reports	Stores damaged equipment records.
Fines	Stores overdue fine information.
Equipment Logs	Stores equipment activity history.

4.2 Relationships and Constraints

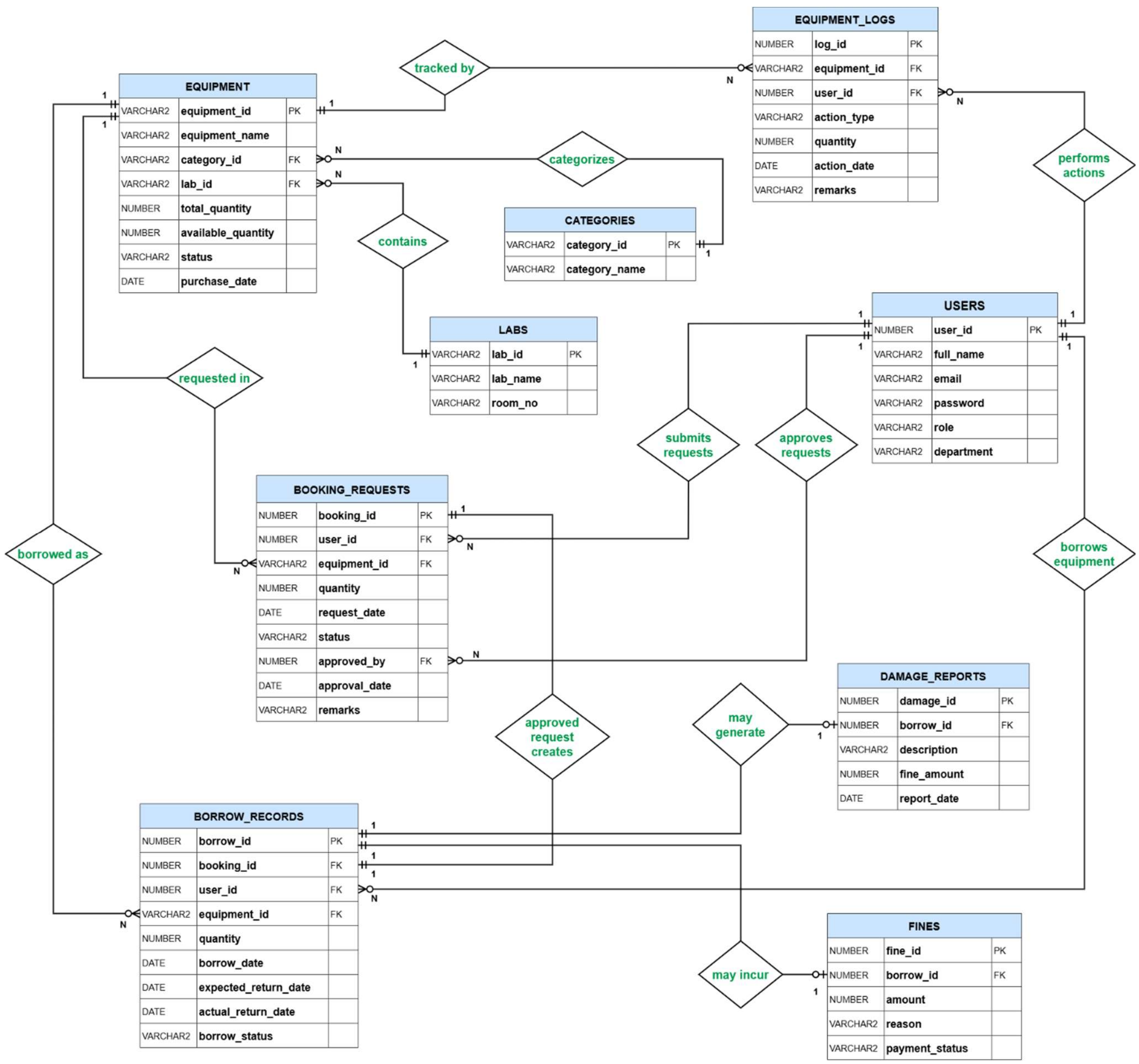

Figure 4.2: Database Entity-Relationship Diagram

Major relationships include:

- Users → Booking Requests
- Users → Borrow Records
- Categories → Equipment
- Labs → Equipment
- Equipment → Booking Requests
- Booking Requests → Borrow Records
- Borrow Records → Damage Reports
- Borrow Records → Fines
- Equipment → Equipment Logs

The database also uses the following constraints:

- Primary Key
- Foreign Key
- NOT NULL
- UNIQUE
- CHECK
- DEFAULT

These constraints ensure valid, secure, and consistent data throughout the system.

4.3 Database Schema

The system uses Oracle Database to manage all application data.

The main database tables are:

- **users** *(user_id, full_name, email, password, role(STUDENT, TEACHER, LAB_ASSISTANT), department)*
- **labs** *(lab_id, lab_name, room_no)*
- **categories** *(category_id, category_name)*
- **equipment** *(equipment_id, equipment_name, category_id, lab_id, total_quantity, available_quantity, status(AVAILABLE, OUT_OF_STOCK, UNDER_MAINTENANCE), purchase_date)*
- **booking_requests** *(booking_id, user_id, equipment_id, quantity, request_date, status(PENDING, APPROVED, REJECTED), approved_by, approval_date, remarks)*
- **borrow_records** *(borrow_id, booking_id, user_id, equipment_id, quantity, borrow_date, expected_return_date, actual_return_date, borrow_status(BORROWED, RETURNED, OVERDUE))*
- **damage_reports** *(damage_id, borrow_id, description, fine_amount, report_date)*

- **fines** *(<u>fine_id</u>, borrow_id, amount, reason, payment_status(PAID, UNPAID))*
- **equipment_logs** *(<u>log_id</u>, equipment_id, user_id, action_type, quantity, action_date, remarks)*

These tables are connected using primary keys and foreign keys to maintain data integrity and support relational database operations.

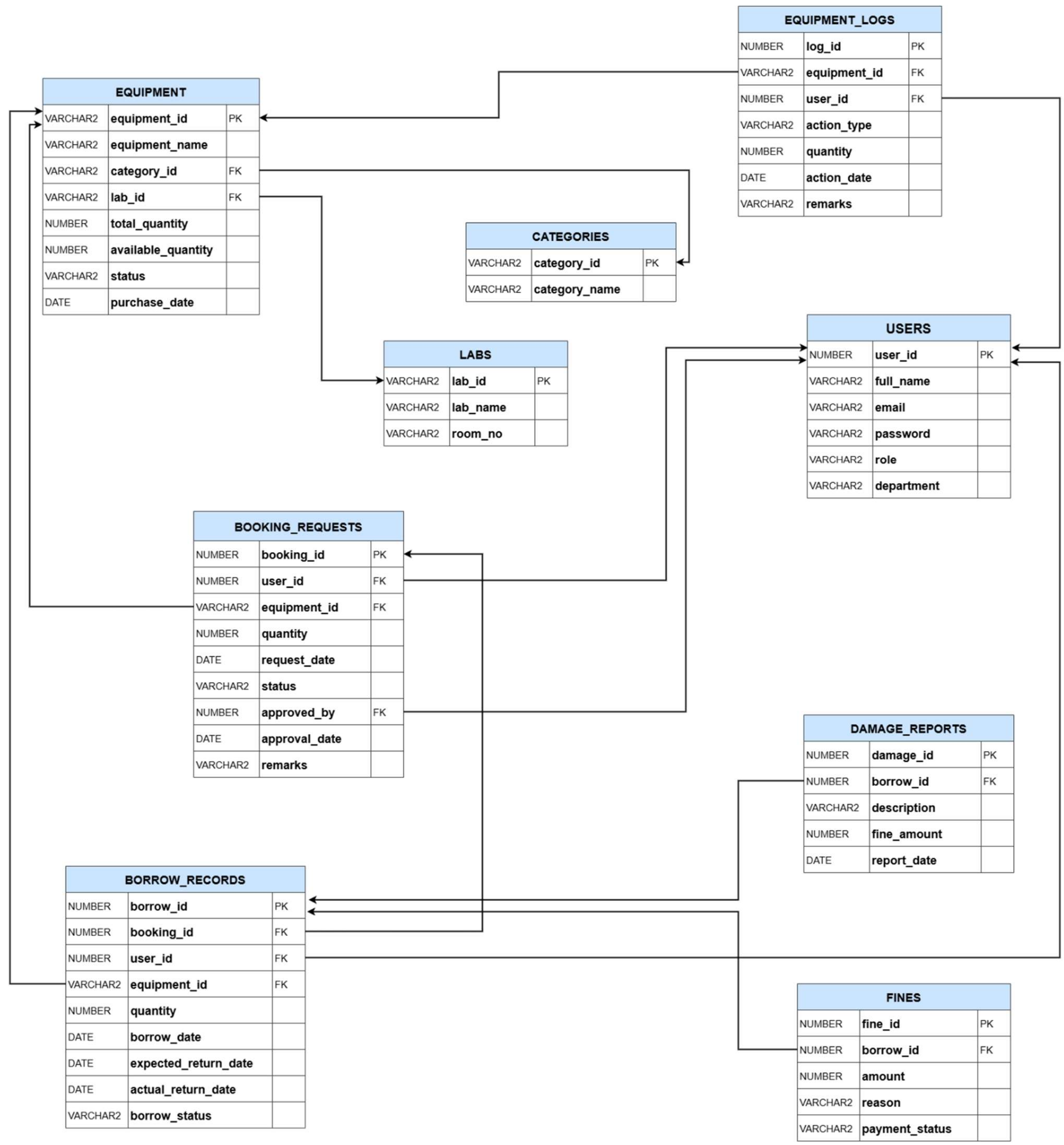

Figure 4.3: Database Schema Diagram

4.4 SQL Operations

The project implements various SQL operations for data management.

DDL Operations

- CREATE TABLE
- PRIMARY KEY
- FOREIGN KEY
- CHECK Constraint
- UNIQUE Constraint

DML Operations

- INSERT
- UPDATE
- DELETE
- SELECT
- Search
- Filter
- Sort

Advanced SQL concepts used in the project include:

- JOIN Operations
- Aggregate Functions
- GROUP BY
- HAVING
- Subqueries

These SQL operations support inventory management, booking, borrowing, fine management, and report generation.

4.5 PL/SQL Components

Oracle PL/SQL is used to automate important database operations.

The project includes:

- Anonymous Blocks
- IF–ELSE Statements
- CASE Statements
- Loops
- Stored Procedures

- User-Defined Functions
- Database Triggers
- Transactions (COMMIT & ROLLBACK)

These components automate business logic such as booking approval, equipment issuing, equipment returning, fine calculation, inventory updates, and activity logging.

5. Implementation

5.1 Development Environment

The system was developed using the following technologies:

Component	Technology
Framework	Laravel 12
Programming Language	PHP
Database	Oracle Database 11g XE
Frontend	HTML, CSS, Bootstrap 5, JavaScript
Database Driver	yajra/laravel-oci8
IDE	Visual Studio Code
Database Tool	Oracle SQL Developer

5.2 Authentication & Role-Based Access

The system uses session-based authentication with three user roles:

- **Lab Assistant** – Full system management.
- **Teacher** – Booking approval and report viewing.
- **Student** – Equipment booking, borrow history, and fine viewing.

Role-based middleware ensures that each user can access only the features assigned to their role.

5.3 Module Implementation

Equipment Management

This module allows the Lab Assistant to manage laboratory equipment. Users can view equipment, while the Lab Assistant can add, update, delete, search, and filter equipment records.

Booking Management

Students can submit booking requests for equipment. Teachers review these requests and approve or reject them based on availability.

Borrow Management

After approval, the Lab Assistant issues the equipment and creates a borrow record. Returned equipment is also managed through this module.

Fine Management

If equipment is returned after the expected return date, a fine is generated. Students can view their fines, while the Lab Assistant manages fine records and payment status.

Reports Module

The Reports module provides summary information about equipment, bookings, borrow records, and fines. These reports help monitor laboratory activities efficiently.

Student & Teacher Management

The Lab Assistant manages student and teacher accounts by adding, updating, searching, and deleting user records. This removes the need for public user registration.

6. Conclusion

The **University Lab Equipment Management System** was developed to simplify and automate laboratory equipment management using **Laravel 12** and **Oracle Database 11g XE**. The system provides a centralized platform for managing equipment, booking requests, borrowing, returns, fines, reports, and user accounts.

Role-based access control ensures that Lab Assistants, Teachers, and Students can perform only their authorized tasks. By replacing manual record keeping with a database-driven solution, the system improves data accuracy, reduces administrative workload, and supports efficient laboratory management.